APPENDIX

M `Galaxy`, A Tidal Interaction Code

`Galaxy` is a program that calculates the gravitational effect of the close passage of two galactic nuclei on a disk of stars. It is adapted from a program written by M. C. Schroeder and Neil F. Comins and published in *Astronomy* magazine. This program is very similar to the one used by Alar and Juri Toomre in 1972 to perform their ground-breaking studies of the effect of violent tides between galaxies.

In the program there are two galactic nuclei of masses M_1 and M_2. They are treated as point masses, and they move under the influence of their mutual gravitational attraction. To speed the calculations, only M_1 is surrounded by a disk of stars, with the stars initially in circular Keplerian orbits. The gravitational influence of the stars is neglected, meaning that they do not affect the motions of the nuclei or one another. There is no dynamical friction, and so the nuclei follow the simple two-body trajectories that were discussed in Chapter 2. One advantage of a non-self-gravitating disk is that results do not depend on the number of stars in the disk. You can experiment, changing the initial conditions by using just a few stars for a faster running time and then increasing the number of stars to see more detail. The stars respond only to the gravitational pull of the two nuclei.

The goal is to calculate the positions of the nuclei and stars through a number of time steps separated by a time interval Δt. Let the positions of the nuclei at time step i be

$$[X_1(i), Y_1(i), Z_1(i)] \quad \text{and} \quad [X_2(i), Y_2(i), Z_2(i)],$$

and let the position of a star be[1]

$$[x(i), y(i), z(i)].$$

Also, let the velocities of the nuclei and the star be

$$[V_{1,x}(i-1/2), V_{1,y}(i-1/2), V_{1,z}(i-1/2)],$$
$$[V_{2,x}(i-1/2), V_{2,y}(i-1/2), V_{2,z}(i-1/2)],$$

and

$$[v_x(i-1/2), v_y(i-1/2), v_z(i-1/2)].$$

[1]The results do not change with the number of stars used, so one star is enough to illustrate the procedure.

The velocities are the average velocities between the present (i) and the previous $(i-1)$ time steps, so

$$v_x\left(i-\frac{1}{2}\right)=\frac{x(i)-x(i-1)}{\Delta t} \tag{M.1}$$

is the x component of the star's velocity. In a similar manner, the x component of the star's acceleration is

$$a_x(i)=\frac{v_x(i+1/2)-v_x(i-1/2)}{\Delta t}. \tag{M.2}$$

Thus the positions and accelerations, which are determined at each time step, "leapfrog" over the velocities, which are determined between time steps.

Given these values of the positions and velocities, the program calculates the x components of the positions and velocities for the next time step in the following way:

1. Find the star's acceleration at the present time step i, using Newton's law of gravity, Eq. (2.11):

$$a_x(i)=\frac{GM_1}{r_1^3(i)}[X_1(i)-x(i)]+\frac{GM_2}{r_2^3(i)}[X_2(i)-x(i)], \tag{M.3}$$

where $r_1(i)$ is the distance between the star and M_1 at time step i,

$$r_1(i)=\sqrt{[X_1(i)-x(i)]^2+[Y_1(i)-y(i)]^2+[Z_1(i)-z(i)]^2+s_f^2}, \tag{M.4}$$

and similarly for $r_2(i)$.

Note that because the nuclei and stars are treated as points, their separations could become very small, even zero (although the conservation of angular momentum makes this rather unlikely). As a result, arbitrarily large values of $1/r_1^3$ and $1/r_2^3$ could cause a numerical overflow and bring a lengthy calculation to an abrupt halt. To avoid this numerical disaster, a *softening factor*, s_f, has been included in the calculations of all separations. This is the smallest separation permitted by the program. Its value is large enough to prevent an overflow, but small enough to have little effect on the overall results.

2. Find the star's average velocity at $i+1/2$,

$$v_x\left(i+\frac{1}{2}\right)=v_x\left(i-\frac{1}{2}\right)+a_x(i)\Delta t. \tag{M.5}$$

3. Find the star's position at the next time step $i+1$, using

$$x(i+1)=x(i)+v_x\left(i+\frac{1}{2}\right)\Delta t. \tag{M.6}$$

4. Find the acceleration of the nuclei at the present time step i, using Newton's law of gravity,

$$A_{1,x}(i) = \frac{GM_2}{s^3(i)}[X_2(i) - X_1(i)] \tag{M.7}$$

and

$$A_{2,x}(i) = \frac{GM_1}{s^3(i)}[X_1(i) - X_2(i)], \tag{M.8}$$

where $s(i)$ is the separation of the nuclei at time step i,

$$s(i) = \sqrt{[X_1(i) - X_2(i)]^2 + [Y_1(i) - Y_2(i)]^2 + [Z_1(i) - Z_2(i)]^2 + s_f^2}. \tag{M.9}$$

5. Find the velocity of the nuclei at $i + 1/2$,

$$V_{1,x}\left(i + \frac{1}{2}\right) = V_{1,x}\left(i - \frac{1}{2}\right) + A_{1,x}(i)\Delta t, \tag{M.10}$$

and similarly for $V_{2,x}(i + 1/2)$.

6. Find the position of the nuclei at the next time step $i + 1$, using

$$X_1(i+1) = X_1(i) + V_{1,x}\left(i + \frac{1}{2}\right)\Delta t, \tag{M.11}$$

and similarly for $X_2(i + 1)$.

The procedure is the same for the y and z components. By repeatedly applying this prescription, it is possible to follow the motions of the nuclei and star(s).

The target galaxy (M_1) is initially placed at rest at the origin. You will be asked to provide the initial position and velocity of the intruder galaxy (M_2), its mass (as a fraction of M_1), and the number of stars around the target galaxy. After each time step, the results are displayed as two graphs showing the positions of the nuclei and stars on the x–y and x–z planes.

You will note in the source code that the program uses a special system of units to speed the calculations. The masses are in units of $2 \times 10^{10}\ \mathrm{M}_\odot$. When the program assigns the target galaxy a mass of 5 in these units, its mass is $10^{11}\ \mathrm{M}_\odot$. The unit of time is 1.2 million years. This is also the value used for the time interval, Δt, so (in these units) $\Delta t = 1$. As a result, Δt does not appear explicitly in the program. (It would just multiply the term involved by 1 and waste computer time.) The unit of distance is 500 pc, and so the unit of velocity is $(500\ \mathrm{pc})/(1.2\ \text{million years}) \simeq 400\ \mathrm{km\ s^{-1}}$. By design, in these units the gravitational constant $G = 1$; thus G does not appear explicitly in the program either.

The source code for `Galaxy`, along with executable versions, is available for download from the resources tab on the following webpage: www.cambridge.org/astrophysics.